

=== CONFIGURACION GNB (1779620100) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 20 MHz bandwidth
To run the srsRAN Project gNB with this config, use the following command:
sudo ./gnb -c gnb.yaml

```
cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF (Open5GS container IP).
    port: 38412
    bind_addr: 0.0.0.0      # CRITICAL: A local IP that the gNB binds to for traffic from the AMF. Must be 0.0.0.0
    supported_tracking_areas:
      - tac: 7
      plmn_list:
        - plmn: "315010"
      tai_slice_support_list:
        - sst: 4            # V2X Slice, as per 3GPP TS 23.501, Table 5.15.2.2-1, Page 199
    inactivity_timer: 7200  # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:
  device_driver: zmq        # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=23.04e6 # ZMQ ports for gNB (tx to
  srate: 23.04e6           # RF sample rate for 20 MHz bandwidth with 15 kHz SCS.
  tx_gain: 75              # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75              # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500         # ARFCN of the downlink carrier (center frequency) for Band 3.
  band: 3                  # The NR band (Band 3, consistent with dl_arfcn 368500).
  channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived (106 PRBs for 15MHz)
  common_scs: 15           # Subcarrier spacing in kHz used for data. (CRITICAL: Must be same as SSB SCS for
  plmn: "315010"          # PLMN broadcasted by the gNB (MCC 315, MNC 010).
  tac: 7                   # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0         # Set search space zero index to match srsUE capabilities
      coreset0_index: 12   # Set search CORESET Zero index to match srsUE capabilities for 20MHz BW
    dedicated:
      ss2_type: common     # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1  # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64       # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64       # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log   # Path of the log file.
  all_level: info          # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false        # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: false       # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true        # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true       # Enable RLC metrics
```

```
    enable_sched: true          # Enable DU scheduler metrics
periodicity:
  du_report_period: 1000      # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779620100) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 23.04e6          # RF sample rate for 20 MHz bandwidth, matches gNB.
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=23.04e6 # ZMQ ports for UE (tx to 2

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3                # NR Band 3, matches gNB.
nof_carriers = 1

max_nof_prb = 106        # Max PRBs for 20 MHz bandwidth with 15 kHz SCS.
nof_prb = 106           # Number of PRBs to use, matches gNB bandwidth.

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage          # USIM algorithm.
opc = 63BFA50EE6523365FF14C1F45F88737D # Operator Code.
k = 00112233445566778899aabbccddeeff # Subscriber Key.
imsi = 315010000000001   # IMSI (MCC 315, MNC 010, MSIN 000000001), matches gNB PLMN.
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn             # Access Point Name.
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER COMPOSE (1779620100) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    env_file:
      - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
    privileged: true
    ports:
      - "9898:9999/tcp"
      # Uncomment port to use the 5gc from outside the docker network
      # - "38412:38412/sctp"
      # - "2152:2152/udp"
    command: 5gc -c open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: 10.53.1.2 # Static IP for Open5GS container.
```

gnb:

```
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  # privileged mode is required only for accessing usb devices
  privileged: true
  # Extra capabilities always required
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    # Access USB to use some SDRs
```

```

- /dev/bus/usb/:/dev/bus/usb/
# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: 10.53.1.3 # Static IP for gNB container.
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ./gnb.yaml # Path to your gNB configuration file.
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0 # CRITICAL: Must be 0.0.0.0 for srsRAN bind UDP.
        metrics:
          autostart_stdout_metrics: true
          enable_json: true
        remote_control:
          bind_addr: 0.0.0.0 # Ensure this bind_addr is also 0.0.0.0.
          enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```